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Reduced sensory attenuation is related to aberrant interoceptive influences on intentional binding in schizophrenia

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Abstract

Aim: Schizophrenia (SZ) is characterized by abnormalities in self-representation, including an aberrant sense of agency (SoA). Experimentally observed abnormalities in force matching and intentional binding tasks suggest that the SoA in SZ is compromised at the implicit level. However, these task effects have been reported independently, leading to inconsistencies that complicate the understanding of SZ pathophysiology.

Methods: Thirty-three patients with SZ and 37 nonclinical (healthy) controls (HCs) performed force matching and intentional binding tasks. The former task assessed sensory attenuation, while the latter evaluated an intentional binding effect, both measuring implicit levels of SoA. We previously observed that the cardiac phase (systole vs diastole) influences intentional binding. Therefore, we investigated the relationship between sensory attenuation and intentional binding, including this "cardiac effect," which refers to the influence of cardiac interoceptive signals.

Results: In SZ, significantly reduced sensory attenuation, a greater but nonsignificant intentional binding effect, and a significant opposite cardiac effect were observed compared with HCs. Sensory attenuation and intentional binding were significantly and positively correlated in the SZ group but not in the HC group. Across all participants, sensory attenuation correlated with the cardiac effect on intentional binding. This effect remained significant in the HC group but not in the SZ group.

Conclusion: Group differences and task associations suggest shared mechanisms and pathophysiology in SZ, potentially involving impaired predictive processes and the failure of sensory attenuation. The cardiac effect may be explained by interoceptive sensory attenuation, and its failure could lead to a disturbed SoA in patients with SZ.

KEYWORDS

interoception, schizophrenia, self, sense of agency, sensory attenuation

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INTRODUCTION

The sense of self is a fundamental conscious experience in humans. In particular, the sense of agency (SoA), the feeling that one is causing or generating an action and its consequences, is proposed as one of the key components of the minimal (biological) self. Altered SoA is argued to underpin the expression of a core feature of schizophrenia (SZ), namely, "self-disturbance."[1] Experimental studies, including our previous research,[2,3] have demonstrated abnormal SoA in patients with SZ, apparent across different paradigms.[4–6] However, a comprehensive interpretation of findings across these independent studies is limited by these different experimental approaches.[6]

Explicit and implicit tests have been applied to assess SoA in experimental settings.[6] Explicit tasks typically deliver manipulated feedback (auditory or visual) following a performed action and then ask a direct question about the participant's SoA. Implicit tasks, such as intentional binding and sensory attenuation tasks (including the force-matching task), probe perceptions and behaviors that index mechanisms that putatively underlie SoA without direct questioning. Differences in findings between explicit and implicit approaches are framed by a hierarchical distinction between the cognitive judgment (explicit) and feeling (implicit) levels of the experience of agency.[4,7,8] However, the relationships between different implicit tasks, especially in SZ, remain underexplored. Among nonclinical participants, one study found no clear relationship between intentional binding and auditory sensory attenuation.[7]

The force-matching task is an implicit task that originated from the observed dampening of the sensory effects of self-generated sensory stimulation (for example, self-tickling) when compared with externally generated stimulation.[9–11] This phenomenon, known as "sensory attenuation," is interpreted as a measure of implicit SoA. A force-matching task was used to assess somatosensory attenuation. Sensory attenuation has been reported for other sensory modalities, including visual and auditory sensations.[6] Proposed mechanisms reportedly include cancellation models and pre-activation theory, both of which involve processes to predict the sensory consequences of self-generated actions.[12] In this context, the comparator model, in which congruence between predicted and actual consequences of action is crucial for the emergence of SoA, provides a compelling account of this phenomenon.[13,14] In SZ, reduced sensory attenuation is observed in the force-matching task and related tasks.[6,10,15] This abnormality is demonstrably independent of illness duration and symptom severity, suggesting that it may be a fundamental pathophysiological trait rather than a direct cause of specific psychotic symptoms, such as hallucinations.[6,12]

The intentional binding task, another widely used implicit task, quantifies how the timings of a self-paced movement and a linked acoustic stimulus (a tone) are perceived.[5] When actions and tones are linked, participants experience the timing of these two events as closer together, a phenomenon known as the intentional binding effect. A stronger intentional binding effect suggests a stronger implicit SoA. The validity of inferring SoA from this task has, nevertheless, been questioned, and recent evidence supports this.[16,17] However, the situation in SZ is somewhat paradoxical since stronger intentional binding has been reported in patients with SZ compared with nonclinical controls.[5] A prevailing explanation is that patients' SoA is largely retrospectively reconstructed based on impaired predictions.[5] In contrast, recent studies highlight the pathological role of weaker binding in conditions like delusions of being controlled,[18] a self-disturbance assessed by semi-structured examinations.[19] These inconsistencies among studies remain unsolved and cannot be solely attributed to the methodological aspects of the paradigm.[6] A more recent model suggests that the SoA (including the intentional binding effect) arises from comparator models (linked to predictions) and hierarchical inferences. Particularly, the latter component works as compensating for impaired predictions in SZ.[20] The inconsistencies among tasks and studies could reflect variations in compensatory hierarchical inferential processes, which could be more readily provoked in the intentional binding task than in the force-matching task.

Our recent study provided additional pathophysiological insights into the intentional binding effect in the context of (cardiac) interoception. The importance of interoception in shaping self-representation has gained increasing recognition.[21–27] Interoception encompasses the body-to-brain signaling, central processing, and (unconscious and conscious) representation of information about the internal physiological state.[28–30] Beneficial effects of interoception are linked to the enhancement of a sense of bodily ownership and agency, more precise action control, and more effective cardiac modulation of cognitions.[26,27,30–33] In particular, healthy individuals with greater interoceptive accuracy, as measured by the heartbeat discrimination task, exhibited stronger intentional binding with greater enhancement during systole.[27] In contrast, dysfunctional interoceptive processing has been demonstrated in SZ.[34–37] Aligning with this, our recent study revealed a disruptive influence of cardiac interoceptive signals, namely the heartbeat, on implicit SoA in SZ.[38] Patients with SZ showed weaker intentional binding during systole than during diastole, whereas healthy individuals demonstrated stronger intentional binding during systole. These findings suggested that abnormalities in central (cardiac) interoceptive processing underlie the pathological expression of SoA.[38]

Interoception encodes basic physiological processes, including cardiac and respiratory dynamics, and it is distinct from exteroception, which deals with sensory information from the external environment, such as the auditory tones used in intentional binding tasks. The intero/exteroceptive classification of aspects of somatosensation, such as pressure sensing in tasks like force matching, is still debated. Events occurring on the skin are located at the anatomical border between self and others. This information may pertain to the internal bodily state; however, somatosensation is not classically included in interoception.[39,40]

This study aimed to achieve a comprehensive understanding of implicit SoA in SZ, including the crucial role of interoception, and address the inconsistency among tasks. We investigated the relationship between intentional binding and sensory attenuation in patients with SZ and a nonclinical comparison group.

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METHODS

Participants

Patients diagnosed with SZ according to the *Diagnostic and Statistical Manual of Mental Disorder*, 5th edition, criteria were recruited from the Shimofusa Psychiatric Medical Center. All patients were clinically stable at the time of testing. The inclusion criteria were as follows: (1) 20–65 years old and (2) ability to perform tasks adequately. The exclusion criteria were (1) patients who were expected to have difficulty performing tasks due to severe mental symptoms, as determined by clinical judgment based on the patients' daily life skills and routine medical interviews; and (2) patients with physical health complications that affect task performance, such as stroke and brain injury. A comparison group of self-reported nonclinical participants was also recruited as age- and sex-matched healthy controls (HCs). The Beck Depression Index-II (BDI) and State–Trait Anxiety Inventory (STAI) were used to clarify the clinical status of the participants. The clinical status of participants was assessed using the Positive and Negative Syndrome Scale (PANSS).[41–43]

General procedure

All participants provided written informed consent before participation. Participants were asked to perform a force-matching task and an intentional binding task in random order. Part of the overall research program highlighting associations between the intentional binding effect and the cardiac cycle has already been published.[38] Therefore, the analysis of intentional binding in this study serves as a sensitivity analysis using the dataset of participants who performed both tasks.

Force-matching task

In this task (Figure 1a), participants first experienced a machine-generated "target force" on their left index fingers and then were asked to replicate this force using either "self"-induced force with their right index finger or a joystick-based generated force (as a surrogate for an "externally" delivered force), referred to as the "matched force." They completed 12 trials in the self-condition, followed by 12 trials in the joystick condition, or vice versa, totaling 24 trials across four sets of three varying forces (2, 3, or 4 *N*), randomly assigned. A higher magnitude of the matched force in the self-condition than that in the joystick condition reflects the degree of sensory attenuation, with a higher degree indicating a greater SoA.

Intentional binding task

In the intentional binding task (Figure 1b),[5] participants observed a clock face on a computer screen and pressed a key with the index finger of their dominant hand following each task condition. There were four conditions and 20 trials in total. The order of the conditions was randomized. In the first and second conditions (action–tone condition), the participants pressed a key, causing a tone 250 ms later, and reported the clock-hand position at the keypress (in the first condition) or tone occurrence (in the second condition). The third (action-only) and fourth (tone-only) conditions served as controls for the first and second conditions and excluded any causal link. In the third condition, the participants pressed a key and reported their clock-hand position at the time of the key press. No tone occurred. In the fourth condition, participants listened to a tone and reported their clock-hand position at its occurrence, without any action being required. The intentional binding effect, which was the main outcome of this task, was defined as the difference between the reported and actual timing of actions and tones.

Additionally, we explored the "cardiac effect" on the intentional binding effect, that is, the difference in the intentional binding effect between systole and diastole. To calculate this effect, the participants wore a pulse oximeter (Nonin Xpod 3012LP) on their left index finger, from which pulse times were recorded to identify relevant (systolic and diastolic) periods within the cardiac cycle. This method of identifying cardiac cycles has already been validated and used in our previous studies.[27,31,38] The standardized cardiac effect was calculated as the raw cardiac effect divided by the intentional binding effect, referring to a purer cardiac effect than the raw effect (uncontrolled effect), as it controls for specific, noncardiac factors (including prospective and retrospective components) that also contribute to the development of SoA.[38]

Statistical analysis

Participant characteristics were compared using *t*-tests, *U*-tests, or χ^2 tests. Clinical data that highly deviated from the normal distribution were described using the median and interquartile range. In the force-matching task, the degree of sensory attenuation was compared between the groups. To address the three target forces, a generalized linear mixed model was used with the degree of sensory attenuation as the dependent variable and group, age, sex, education, heart rate, STAI-State, and BDI as independent variables. To address the nonnormal distribution of the degree of sensory attenuation, these data were transformed using the Yeo–Jonson transformation and then entered into the model.[44] The intentional binding effect and standardized cardiac effect were also compared between groups, using general linear models to compare these effects between the groups with the same independent variables. Because the standardized cardiac effect is estimated by dividing the raw cardiac effect by the overall intentional binding effect, the negative values of the intentional binding effect are problematic. To ensure consistent meaning of the calculated effect and reduce the risk of heterogeneity,[38,45] we excluded participants who showed no intentional binding effect from our analysis. This selection follows our previously published study.[38] Within the patient group, to explore the relationship with clinical status, Spearman correlation analyses and Wilcox tests were conducted with various clinical

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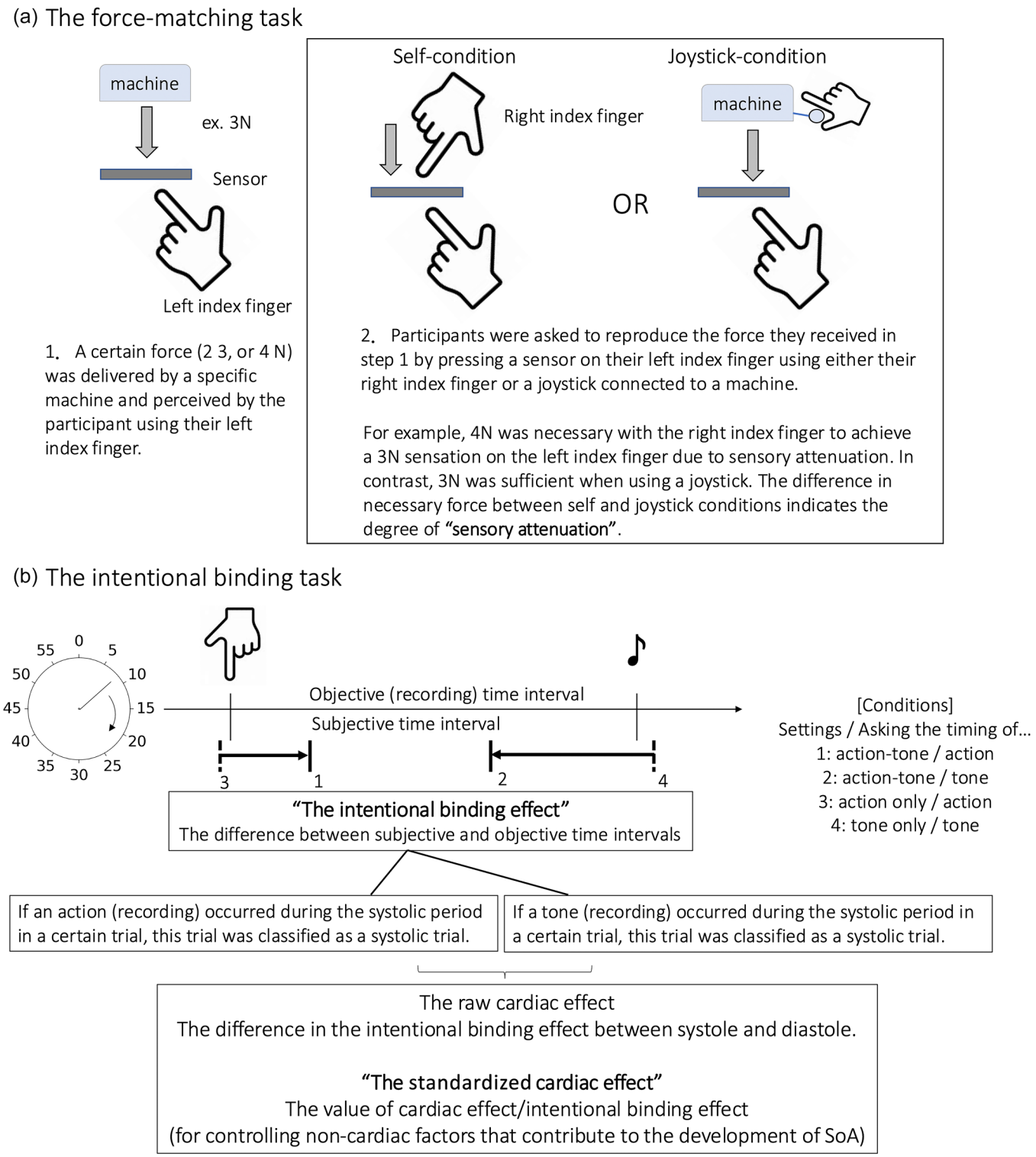

FIGURE 1 (a) In the force-matching task, a higher magnitude of the matched force in the self-condition compared to the joystick condition reflects the degree of sensory attenuation, with a higher degree indicating greater sense of agency (SoA). (b) In the intentional binding task with cardiac monitoring, the intentional binding effect and the standardized cardiac effect were estimated.

information, including PANSS scores, the number of hospital admissions, antipsychotic dosage, duration of illness, and patient status (inpatient or outpatient).

This study aimed to quantify the relationship between sensory attenuation and intentional binding/standardized cardiac effects. To address potential confounders and the relatively small number of participants in each group, stepwise selection was applied to a general linear model. The model used the average sensory attenuation among target forces as the dependent variable and included the intentional binding effect or standardized cardiac effect as

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TABLE 1 Characteristics of participants.

	Schizophrenia *N* = 33	Healthy control *N* = 37	*p* value
General information			
Age, years, mean ± SD	41.9 ± 10.9	42.4 ± 10.0	0.82
Male, *N* (%)/Female, *N* (%)	14 (42)/19 (58)	16 (43)/21 (57)	1.00
Education, years, mean ± SD	13.3 ± 2.1	17.1 ± 2.1	<0.01
BMI, kg/m^2, mean ± SD	24.1 ± 4.1	23.5 ± 5.0	0.54
HR, bpm, mean ± SD	83.0 ± 19.3	73.3 ± 12.0	0.02
BDI, mean ± SD	12.8 ± 11.1	6.5 ± 6.1	0.01
STAI-Trait, mean ± SD	45.7 ± 11.2	40.4 ± 8.0	0.03
STAI-State, mean ± SD	47.7 ± 11.4	39.3 ± 8.9	<0.01
Clinical information			
Duration of illness, years, median [IQR]	19.0 [13.0]	–	
Number of hospital admissions for hallucination and delusion, median [IQR]	2.0 [3.0]	–	
Outpatient, *N* (%)/Inpatient, *N* (%)	14 (42)/16 (48)	–	
Drug dosage, CP-equivalent mg, median [IQR]	750.0 [550.0]	–	
PANSS–Positive, median [IQR]	10.0 [7.0]	–	
PANSS–Negative, median [IQR]	12.0 [9.0]	–	
PANSS–General, median [IQR]	28.0 [13.0]	–	
PANSS–Total, median [IQR]	52.0 [27.0]	–	

Abbreviations: BMI, body mass index; BDI, beck depression inventory; CP, chlorpromazine; HR, heart rate; IQR, interquartile range; PANSS, positive and negative syndrome scale; SD, standard deviation; STAI, State-Trait Anxiety Inventory.

independent variables, which were not intended for elimination in the stepwise selection. The initial models for the total sample and the HC group included the same independent variables as the aforementioned models for group comparison. The group effect and the interaction between the group and the intentional binding effect/standardized cardiac effect were included only in the model for the total sample and were not targeted for selection. The initial model for SZ also included clinical information that was the same as that used when exploring its relationship with task performance. Statistical significance was set at $p < 0.05$ after correcting for multiple comparisons using the false discovery rate (FDR). Statistical analyses were performed using R (4.3.2 or later).[46]

RESULTS

Characteristics of the participants

The 39 patients with SZ and 39 HCs performed force-matching and intentional binding tasks. Of these, one HC and six patients with SZ were excluded from our analysis because of poor performance determined by low correlation ($r^2 < 0.04$) between the matching and corresponding target forces in either condition.[10] One HC was excluded based on post-task conversations indicating inadequate task understanding. Finally, data from 33 patients and 37 HCs were analyzed for the force-matching task and intentional binding effect (Table 1). All patients were treated with antipsychotic drugs at a CP-equivalent dose of median [IQR]: 750.0 [550.0] mg. Thirteen patients also received sleep medications, including benzodiazepines and orexin receptor antagonists. Only five patients used hypnotic drugs during the daytime or long-acting hypnotic drugs. Two patients were prescribed valproic acid as a mood stabilizer. Four patients were prescribed anticholinergic drugs to manage extrapyramidal symptoms.

Of these, some participants were excluded because of heartbeat recording failure and negative values of intentional binding effects as aforementioned reasons. Therefore, the datasets of 28 patients and 30 HCs were analyzed for the force-matching task and its relationship with standardized cardiac effects on intentional binding.

Group differences in the force-matching task

Our model revealed a significant group difference in the degree of sensory attenuation ($B = -1.45$, $p = 0.025$) after controlling for age,

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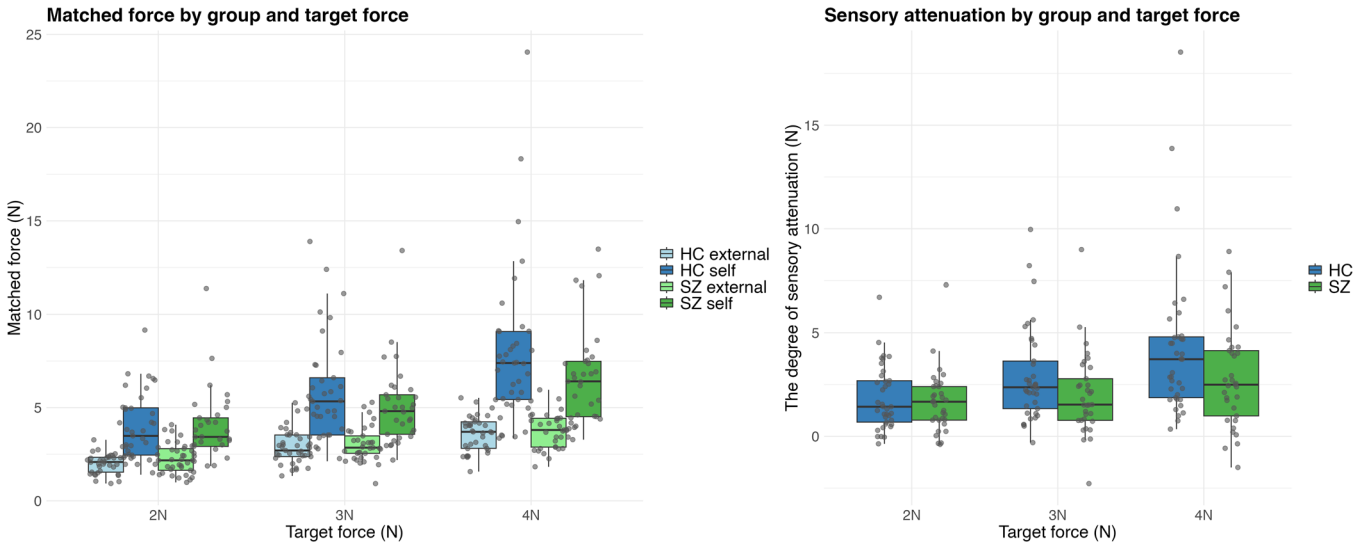

FIGURE 2 Left: The required force to match the target force was greater in the self-condition than in the joystick (external) condition, regardless of group. Right: The degree of sensory attenuation was significantly greater in healthy controls (HCs) than in patients with schizophrenia (SZ), indicating blunted sensory attenuation in SZ.

sex, education, heart rate, anxiety, and depression (these factors were not significant), indicating blunted sensory attenuation in patients with SZ compared with HCs (Figure 2). In a post-hoc analysis of the conditions, significant group differences in 3 *N* and 4 *N* remained after FDR correction ($p = 0.043$ and 0.043, respectively).

Within patients, the overall degree of sensory attenuation was not significantly associated with PANSS (Positive [$\rho = 0.16$, $p = 0.37$], Negative [$\rho = -0.08$, $p = 0.68$], General [$\rho = -0.05$, $p = 0.78$]), the number of hospital admissions for delusions and hallucinations ($\rho = 0.03$, $p = 0.86$), antipsychotic dosage ($\rho = -0.10$, $p = 0.59$), or duration of illness ($\rho = -0.09$, $p = 0.63$), but had a trend of reduction in inpatients compared with outpatients ($W = 185$, $p = 0.07$).

Group differences in the intentional binding task

The overall average intentional binding effect was greater in the SZ group than in the HC group; however, the difference was not significant ($B = 16.5$, $p = 0.65$) after controlling for age, sex, education, heart rate, anxiety, and depression, none of which were significant factors (Figure 3, left). In contrast, the standardized cardiac effect on the intentional binding effect was significantly different between the SZ and HC groups, with opposite effects of cardiac (systole vs diastole) on the intentional binding effect. Specifically, cardiac systole disrupted intentional binding in the SZ, in contrast to enhanced intentional binding in the HC ($B = -0.91$, $p = 0.001$) (Figure 3, right). No other factors were found to be significant in the model.

Within patients, the intentional binding effect did not significantly correlate with PANSS (Positive [$\rho = 0.05$, $p = 0.78$], Negative [$\rho = 0.05$, $p = 0.78$], General [$\rho = -0.03$, $p = 0.87$]), the number of hospital admissions for delusions and hallucinations ($\rho = -0.04$, $p = 0.83$), antipsychotic dosage ($\rho = -0.12$, $p = 0.51$), duration of illness ($\rho = 0.21$, $p = 0.24$), or treatment settings ($t = 0.36$, $p = 0.72$). The standardized cardiac effect was not significantly associated with PANSS (Positive [$\rho = 0.19$, $p = 0.92$], Negative [$\rho = 0.09$, $p = 0.63$], General [$\rho = -0.05$, $p = 0.80$]), the number of hospital admissions for delusions and hallucinations ($\rho = -0.32$, $p = 0.09$), antipsychotic dosage ($\rho = -0.20$, $p = 0.29$), duration of illness ($\rho = 0.17$, $p = 0.37$), or treatment settings ($W = 98$, $p = 0.58$).

Relationship between these tasks

The relationships between these tasks are illustrated in Figure 4. In our stepwise models using the total intentional binding effect (Table 2), the degree of sensory attenuation was significantly associated with the degree of the total intentional binding effect in the SZ ($B = 0.007$, $p = 0.013$) but was insignificant in the models using the total sample or HC. The significance observed in the SZ group persisted after FDR correction ($p = 0.039$).

Furthermore, in models using the standardized cardiac effect, the degree of sensory attenuation was significantly associated with the degree of standardized cardiac effect across the total sample ($B = 1.170$, $p = 0.009$) and HCs ($B = 1.170$, $p = 0.010$), but not in patients with SZ (Table 2). These effects remained significant after the FDR correction (0.015 and 0.015, respectively).

DISCUSSION

To our knowledge, this is the first study to reveal the relationship between sensory attenuation and intentional binding/standardized cardiac effects in patients with SZ. We observed a positive relationship between sensory attenuation and the intentional binding

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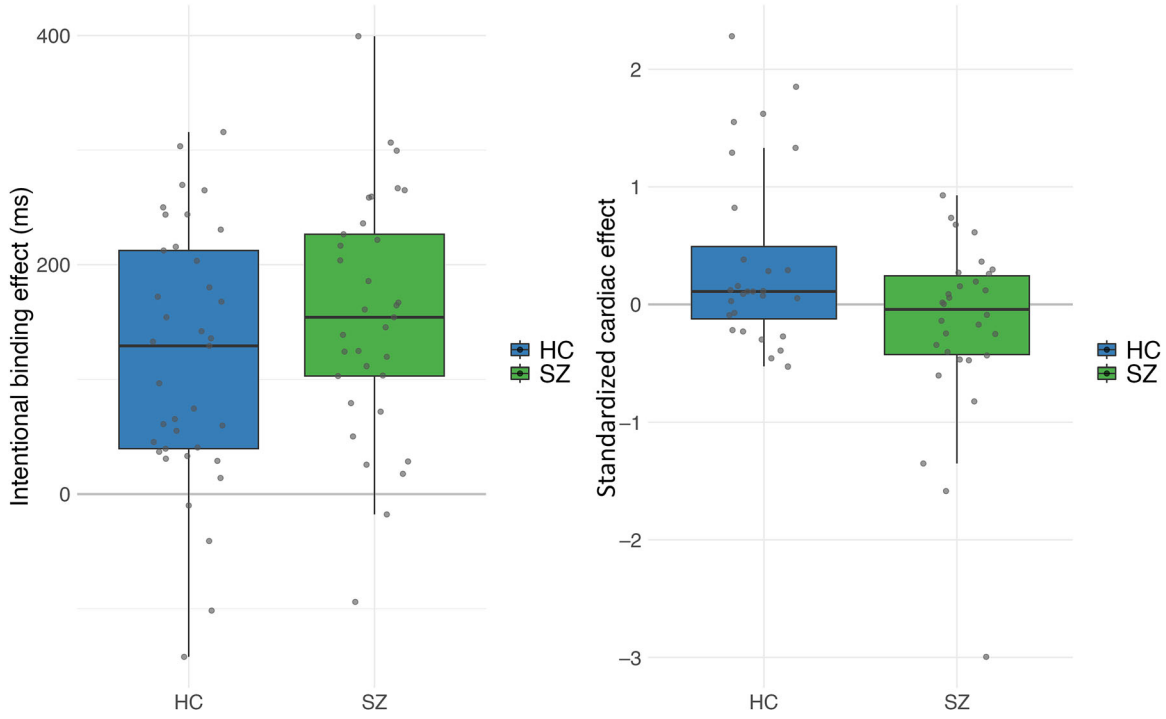

FIGURE 3 Left: While a greater average of the intentional binding effect was found in patients with schizophrenia (SZ) than in healthy controls (HCs), the difference did not reach significance, particularly after controlling for potential confounding factors. Right: The standardized cardiac effect on the intentional binding effect was significantly different between SZ and HC groups, with opposite effects of cardiac systole on sense of agency (SoA). Specifically, cardiac systole disrupted SoA in SZ, contrasting with the enhanced SoA in HC.

effect in patients with SZ but not in nonclinical participants, suggesting a shared pathophysiology affecting these tasks in SZ. Additionally, we observed a positive relationship between sensory attenuation and standardized cardiac effects across the total sample and in nonclinical participants, perhaps indicating a unique mechanism related to cardiac interoceptive signals that may have been obscured by heterogeneity among patients with SZ when only SZ data were analyzed. These findings cannot be explained by individual inaccuracy or cognitive deficits in SZ because a smaller degree of sensory attenuation and a weaker intentional binding effect paradoxically mean increased accuracy or correspondence between self-generated and other-generated action-induced perceptions, as well as between subjective and objective temporal judgments.

We begin by discussing the observed group differences in task performance. The force-matching task revealed significantly reduced (blunted) sensory attenuation in SZ, consistent with findings across different methodologies,[6,12] including a force-matching task.[10] This attenuation, most pronounced at higher target forces, aligns with Weber–Fechner and Stevens's psychophysical laws.[47] The lack of association with psychotic symptoms remains in line with these studies, supporting the notion that reduced sensory attenuation is not a sufficient explanatory mechanism for psychotic symptoms but rather a broader pathophysiological vulnerability apparent at any stage of SZ, including the prodromal phase.[12]

Our findings from the intentional binding task served as a sensitivity analysis in our previous study.[38] A nonsignificant trend toward a stronger intentional binding effect in SZ, particularly after controlling for potential confounding factors, suggests that stronger binding in SZ is not a definitive phenomenon, highlighting the inconsistencies among existing studies.[6] This effect may be affected by various factors, including compensatory hierarchical inferential processes based on their impaired prediction.[6,20] Furthermore, we revealed a significant group difference in the influence of cardiac interoceptive signals on the intentional binding effect. Specifically, binding was weaker during systole compared with diastole in the SZ, whereas it was stronger during systole in the HC. This influence can be considered a purer cardiac effect, as it was calculated based on a comparison between the systolic and diastolic periods, with adjustments for the total binding effect. These processes may minimize the impact of specific, noncardiac factors, including inferential processes.[38] This aligns with previous studies on healthy participants and evidence of impaired interoceptive processing in SZ.[26,27,34–37]

The observed link between reduced sensory attenuation and weak binding in SZ implies a shared pathophysiology between these tasks in patients with SZ, possibly involving impaired predictive mechanisms. According to most agency models, including the comparator model, impaired prediction explains the reduced sensory attenuation.[6,12] This explanation and our findings on this link suggest that impaired prediction may contribute to weaker binding. This is consistent with the pathological role of weaker binding in conditions such as delusions of being controlled and self-disturbance assessed by semi-structured examinations.[18,19] However, the paradoxical

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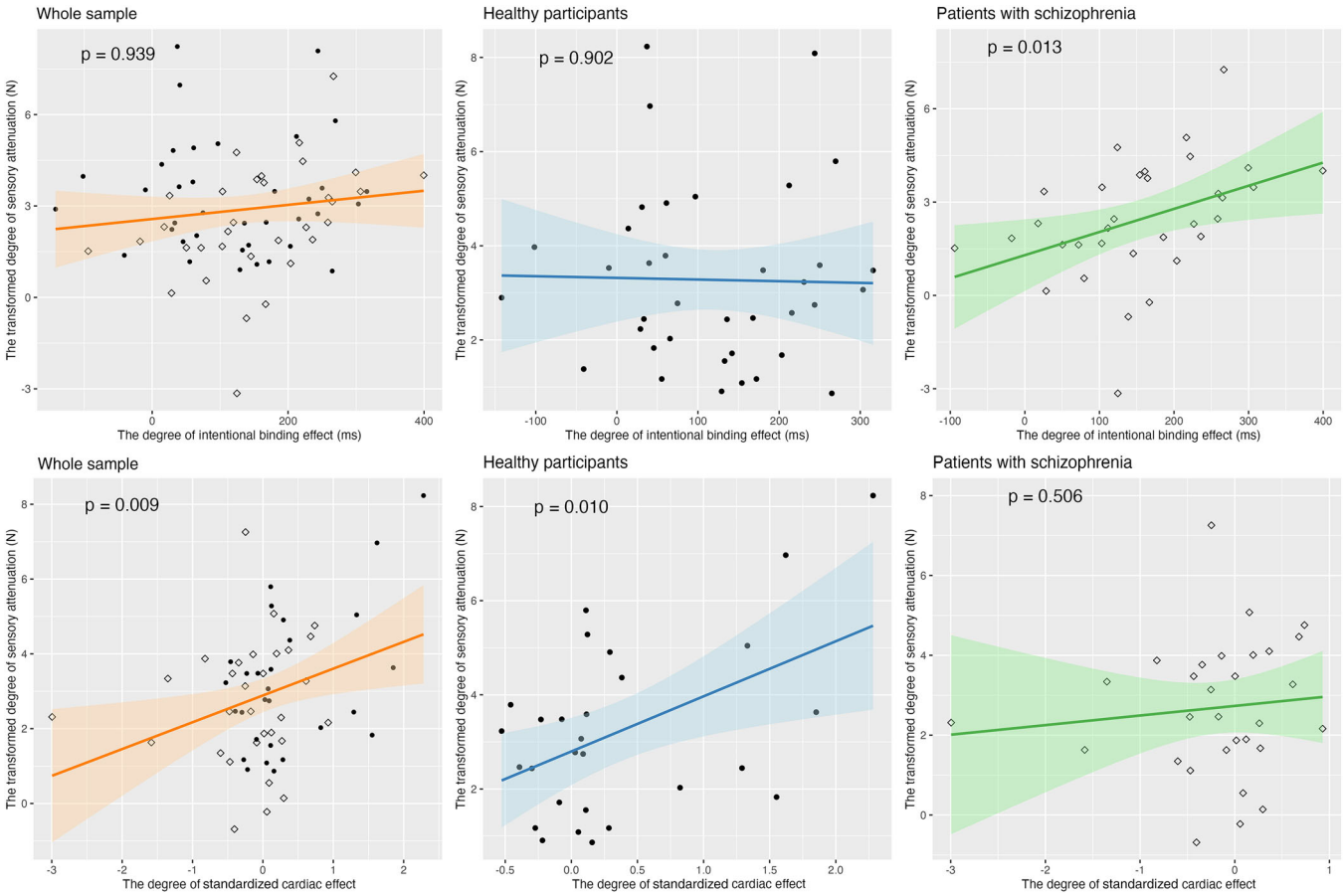

FIGURE 4 The intentional binding effect was significantly associated with sensory attenuation in patients with schizophrenia (SZ) but this was not significant in the models using the total sample or healthy controls (HCs). The standardized cardiac effect was significantly correlated with sensory attenuation in the model using the total sample and in HC but not in patients with SZ. The degree of sensory attenuation was transformed using the Yeo–Jonson transformation and plotted.

TABLE 2 The relationship between the level of sensory attenuation and the intentional binding effect, including the standardized cardiac effect.

Models	*B*	Standard error	*t*-value	*p* value	Other remaining factors in final models
Intentional binding effect					
Total sample	0.0002	0.003	0.076	0.939	Group ($p = 0.012$), age ($p = 0.08$), sex ($p = 0.16$), interaction ($p = 0.10$)
HC	−0.0004	0.003	−0.124	0.902	-
SZ	0.007	0.003	2.655	**0.013**	Age ($p = 0.07$), inpatient ($p = 0.04$)
Standardized cardiac effect					
Total sample	1.170	0.430	2.721	**0.009**	Group ($p = 0.89$), interaction ($p = 0.12$)
HC	1.170	0.421	2.782	**0.010**	-
SZ	0.276	0.409	0.674	0.506	age ($p = 0.06$), inpatient ($p = 0.08$)

Note: The transformed degree of sensory attenuation was set as a dependent variable in these stepwise models. All initial models included age, sex, education, heart rate, anxiety and depression. Initial models for SZ also included clinical information, such as PANSS scores, the number of hospital admissions, antipsychotic dosage, duration of illness, and patient status (inpatient or outpatient). Models using total sample also included a group factor and its interaction with the intentional binding effect/standardized cardiac effect. To address multiple factors, a stepwise selection was applied. The intentional binding effect/standardized cardiac effect, the group factor, and their interaction were included final models.

Abbreviations: PANSS, Positive and Negative Syndrome Scale; HC, healthy control; SZ patients with schizophrenia.

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trend toward stronger binding in the SZ has also been (and was originally) reported and may involve retrospective and inferential processes.[5] Our finding of a nonsignificant group difference further implies the involvement of such compensatory processes. This combination of findings suggests that the binding effect in the SZ could comprise two components: a quantitatively influenced weaker direction driven by impaired predictions and a qualitatively or fixedly influenced stronger direction driven by compensatory inferential processes in the intentional binding task. This interplay results in a persistent correlation, even with a compensatory binding effect. The absence of a similar relationship in nonclinical participants may reflect normal variation within this population, reducing our sensitivity to detect such a relationship, or it may result from differences in cross-modal sensory prediction within each task, that is, somatosensory versus auditory predictions. In SZ, these impairments occur at a pathological level, which allows for a correlation reflecting general prediction impairment, regardless of modality.

The sensorimotor prediction model in intentional binding can be extended to understand the standardized cardiac effect through interoceptive prediction and the sensory attenuation of interoceptive signals.[48,49] Unlike action-related predictions, higher-order visceromotor interoceptive predictions of involuntary heartbeats are less direct. However, the efferent visceromotor drive, which manifests as states of cardiovascular arousal, is linked to physical activity and action-ready emotional states that provide exteroceptive and interoceptive feedback. The predictive coding model, when applied to interoception, offers a framework for self-representation and agency.[48,50,51] The model posits that the brain consistently compares predicted and actual visceroseensory signals and updates its predictions to minimize discrepancies. In this context, similar to sensorimotor predictions, we speculated that sufficiently precise interoceptive predictions attenuate incoming cardiac interoceptive signals to mitigate the detrimental (interfering) impact on action control and sensory processing in other modalities to optimize brain function while simultaneously engendering SoA. Given the significant relationship between sensory attenuation and the standardized cardiac effect in HC, rather than intentional binding, sensorimotor predictions may be more closely linked to interoceptive predictions than auditory exteroceptive predictions.[39] It might also be attributable to the similarity between sensorimotor and interoceptive predictions in lifelong learning about one's own body, which differs from experimental set tones.[52]

The detrimental effect of cardiac interoceptive signals on implicit SoA in SZ may be explained by the failure of interoceptive attenuation arising from a mismatch between interoceptive prediction and upcoming cardiac interoceptive signals. The positive but insignificant relationship observed in the SZ group implies heterogeneity in this population. The impaired sensorimotor prediction assessed in the force-matching task is likely linked to impaired interoceptive prediction but not directly linked to deficits in afferent interoceptive signaling. Both aspects could be compromised in SZ, eventually leading to a mismatch regardless of which is compromised. This effect may also obscure the clear relationship with SZ. Anatomical and functional abnormalities within the insular cortex,[53,54] a central neural substrate for interoception that is engaged in predictive processing,[55,56] have been identified in SZ. Using electroencephalography, we previously demonstrated differences in heartbeat-evoked potentials (a neural signature of cardiac afferent signaling in the brain) between patients with SZ and HC, potentially indicating a failure to attenuate interoceptive prediction error.[57] In another study, we showed that interoceptive prediction error indexed by the discrepancy between subjective belief and objective task performance regarding interoception was significantly correlated with positive symptoms of SZ.[34] Furthermore, the attenuation of interoceptive signals may affect the integration of interoception and exteroception. In our recent study, the cardiac effect in healthy individuals was significantly correlated with interoceptive accuracy assessed by the heartbeat discrimination task, indexing the capability for such cross-modal integration.[27] In a predictive coding framework related to this account, interoceptive signals and their predictions contribute to SoA via their integration with predicted and actual sensory feedback from action.[24] Conversely, a failure of interoceptive attenuation could disrupt this integration process and lead to aberrant SoA in SZ.

This study had some limitations. First, acute patients with severe symptoms were not included, possibly leading to insufficient power to detect associations between task performance and psychotic symptoms, thereby limiting the generalizability of our findings. Second, compared with previous studies on intentional binding and force-matching tasks,[10,11,27] the number of trials was restricted to ensure that participants, especially patients, could perform the tasks feasibly, as indicated by our preliminary study. A greater number of participants were included to compensate for this limitation; however, this may have reduced the statistical power of our analysis. Third, the "classic" intentional binding task does not provide insights into the prospective and retrospective components of SoA.[5] Moreover, only implicit measures of SoA were evaluated; explicit levels of SoA were not examined, which restricts a comprehensive understanding of SoA. Fourth, although CP-equivalent dosage did not correlate with our outcomes, antipsychotic drugs have diverse and overlapping effects. Furthermore, some participants were treated with a broad range of medications, including benzodiazepines and mood stabilizers. This complexity is particularly pronounced in patients taking multiple medications, thereby making it difficult to differentiate the specific effects of each medication. Therefore, the potential effect of medication on group differences cannot be ruled out. Finally, cognitive dysfunction in patients with SZ remains a limitation. To address this, the task-specific cognition required to conduct the tasks was adjusted through comparisons with the control conditions for each task. This adjustment process may minimize the impact of cognitive dysfunction on task performance; however, it does not fully eliminate its influence.

In conclusion, the association between the two tasks indicates the likelihood of a shared pathophysiology involving aberrant predictive mechanisms in SZ. The link between sensory attenuation and the standardized cardiac effect on intentional binding suggests that the successful attenuation of cardiac interoceptive signals based on interoceptive prediction may contribute to a beneficial cardiac effect

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on implicit SoA. Conversely, the failure of interoceptive attenuation may lead to a detrimental cardiac effect in SZ, where predictions and afferent signals may result in mismatches and unresolved prediction errors. Further investigation is required; however, impaired interoceptive prediction and failure of interoceptive attenuation may underpin vulnerability to disturbances in the SoA, potentially triggering psychotic symptoms.

AUTHOR CONTRIBUTIONS

Akihiro Koreki: Conceptualization; methodology; software; validation; formal analysis; investigation; resources; data curation; writing—original draft; writing—review and editing; visualization; project administration; funding acquisition. **Yuri Terasawa:** Validation; writing—review and editing; supervision. **Hiroki Oi:** Validation; writing—review and editing; supervision. **Takuya Mouri:** Methodology; software; validation; investigation; resources; data curation; writing—review and editing. **Aki Takenaka:** Methodology; software; validation; investigation; resources; data curation. **Teruki Koizumi:** Data curation; writing—review and editing. **Mitsumoto Onaya:** Resources; data curation; writing—review and editing. **Mahinda Yogarajah:** Methodology; validation; investigation; supervision; writing—review and editing. **Hugo Critchley:** Validation; writing—review and editing, supervision. **Atsuo Nuruki:** Conceptualization; methodology; software; validation; investigation; resources; data curation; writing—review and editing; supervision.

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CONFLICT OF INTEREST STATEMENT

None of the authors have any conflict of interest to declare.

DATA AVAILABILITY STATEMENT

The datasets generated during the current study will be made available by the corresponding author upon reasonable request and are subject to regulatory approval.

ETHICS APPROVAL STATEMENT

This study was approved by the Ethics Committee of Shimofusa Psychiatric Medical Center (202112001) and conducted in accordance with the ethical guidelines set forth by the Declaration of Helsinki.

PATIENT CONSENT STATEMENT

All participants gave written informed consent prior to participation.

CLINICAL TRIAL REGISTRATION

This study has been registered at UMIN (UMIN000047287).

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